Curriculum Vitae

Nov 2026

Personal Info

Name:	Boyuan Zhao
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Education

10/2021- 09/2025	PhD in Mathematics, **University of St Andrews**, Supervised by Mike Todd and Kenneth Falconer Dissertation: *Exponential mixing and limit theorems in dynamical systems*.
09/2020–08/2021 Distinction (GPA 17.3/20)	MSc Mathematics **University of St Andrews** Dissertation: *On rotation number and unique minimal sets of circle homeomorphisms*. Advisor: Mike Todd
09/2017–08/2020 First Class Honours	BSc Mathematics with Modern Languages **University College London**

Employment

03/2026-	Postdoc - Beijing International Centre for Mathematical Researches (BICMR), Peking University, China. I am organising the Postdoc Seminar for BICMR, see BICMR Seminar Calendar.

Research Interests

I am interested in various topics and problems in ergodic theory and dynamical systems. More precisely,

- Countable Markov shifts and Strong positive recurrence theory,
- Dynamical Borel-Cantelli lemmas and one-dimensional dynamics,
- Extreme Value Theory, Recurrence events, Waiting time distributions,
- Multifractal spectrum for maps modelled by countable Markov shifts.

Research Output

1. *On record behaviours of returns for interval maps*, preprint, 2026. arxiv (31 pages)
2. (with Godofredo Iommi and Mike Todd) Convergence of the Birkhoff spectrum for non-integrable observables, (joint with Godofredo Iommi and Mike Todd) Math. Z. **314**(48) (2026). link
3. *Almost sure cover times for dynamical systems with ψ-mixing property*, Ergod. Theory Dynam. Syst., **46**(1) (2026) 320-336. link.
4. (with Mike Todd) *Countable Markov Chain with Exponential Mixing*. (2024) preprint, arxiv (24 pages).
5. *Closest distance between iterates of typical points* . Discrete Contin. Dyn. Syst. 44(8), (2024) 2252-2279. link

Teaching

I am teaching Linear Algebra (C) at Peking University during the Fall semester 2026.

For each semester during my PhD, I have been a tutor and an example- class demonstrator for the following modules:

- MT1002 Mathematics
- MT1003 Pure and Applied Mathematics
- MT2501 Linear Algebra
- MT2502 Analysis

Conferences Attended and Talks

Oct 2026	1st Hebei International Conference on Dynamical Systems, @ Hebei Normal University, China.
Aug 2026	Dynamical systems seminar, @SIMIS (Fudan University) *Talk title: Convergence rate of the Birkhoff spectrum for non-integrable observables*.
Mar 2026	Frontiers in Dynamical Systems,
Jun 2025	BMC & BAMC 2025, @ University of Exeter.
Jun 2025	Beyond Uniform Hyperbolicity II, ICTP
Mar 2025	Scottish Dynamics Network meeting, @ Herriot-Watt University *Talk title: Almost sure cover times in dynamical systems*.
Jul 2024	Workshop on Dynamical Systems: Geometric and Statistical Properties University of Warwick
Jun 2024	BMC 2024, @ University of Manchester *Talk title: Countable Markov shifts with exponential mixing*.
Dec/2023	Workshop on Probabilistic Methods in Dynamics @ Universidade do Porto *Talk title: Maximal length of subsequence matching and Renyi entropy*.

I have also given four talks at the University of St Andrews analysis seminar. See GROUP PAGE for titles and abstract.

Scholarship and Funding

10/2021-10/2025 China Scholarship Council Programme

Other Interests and Skills

I can speak Spanish at B2 level and Portuguese at B1.

I am an Irish & Scottish folk musician. Video. I have been the vice president of the University of St Andrews folk and traditional music society, and the coordinator of our performance/ceilidh band.

I have competed 400m for University of St Andrews in the SSS indoor championships. I am now a distance/trail runner.